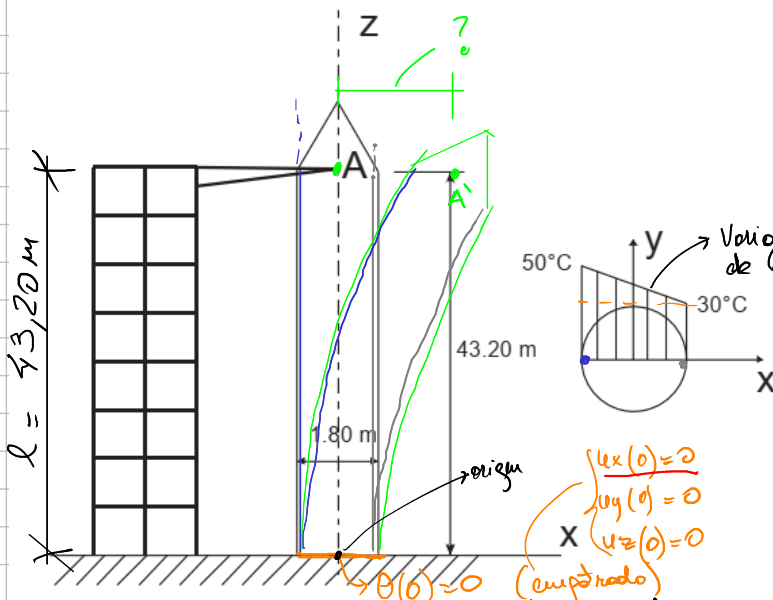


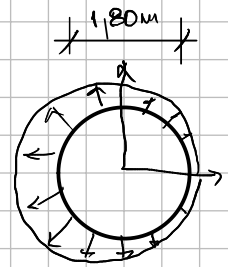
Pequeñas deformaciones

$$\alpha = 10^{-5} \frac{\text{cm}}{\text{cm}^\circ\text{C}}$$

$$\epsilon_{ij} \alpha \Delta T = 20^\circ\text{C} \cdot 1 \times 10^{-5} \frac{\text{cm}}{\text{cm}^\circ\text{C}} = 2 \times 10^{-4} \frac{\text{cm}}{\text{cm}}$$



Variación lineal de la temperatura dentro del cuerpo



$$\begin{cases} u_x(0) = 0 \\ u_y(0) = 0 \\ u_z(0) = 0 \end{cases}$$

$\theta(0) = 0$ (empotrado)

con respecto a una temperatura de referencia

$$\epsilon_{ij} = \alpha \Delta T \cdot \mathbb{I}_{ij}$$

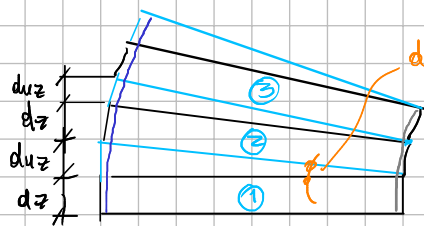
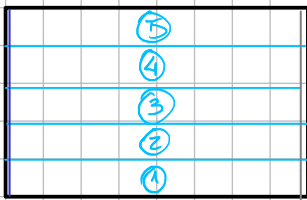
$$\epsilon_{ij} = \alpha \Delta T \delta_{ij}$$

Dada deformación uél es el desplazamiento

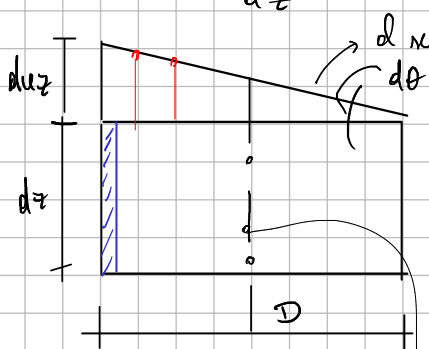
$$\epsilon_{xx} = \int \frac{\partial u_x}{\partial x} \Rightarrow u_x = f(x, y, z) + C(y, z)$$

$$\epsilon_{zz} = \int \frac{\partial u_z}{\partial z}$$

$$u_x = \alpha \Delta T (x - x_0) + C(y, z) \downarrow \frac{\alpha \Delta T}{D} \cdot \frac{z^2}{2}$$



$$\epsilon_{zz} = \frac{du_z}{dz} = (\alpha \Delta T \mathbb{I}_{zz})_{zz} = \alpha \Delta T \cdot 1 \Rightarrow du_z = \alpha \Delta T dz$$



a no lineal la variación del incremento de temperatura

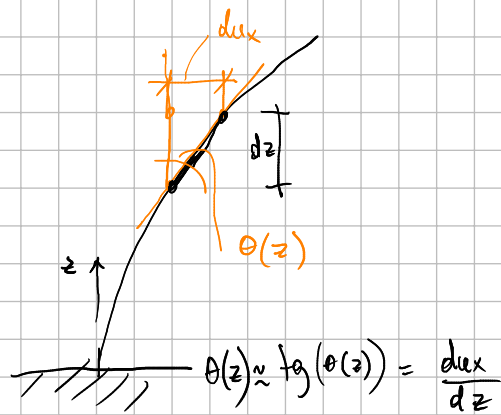
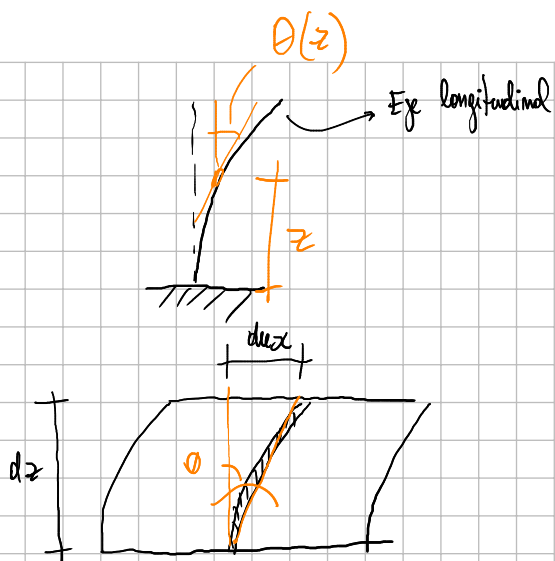
$$\tan(d\theta) \approx d\theta = \frac{du_z}{D} = \frac{\alpha \Delta T dz}{D}$$

$$\int_0^z d\theta = \int_0^z \frac{\alpha \Delta T}{D} dz$$

$$\theta(z) - \theta(0) = \frac{\alpha \Delta T}{D} \int_0^z dz = \frac{\alpha \Delta T}{D} (z - 0) = \frac{\alpha \Delta T}{D} z$$

$$\theta(z) = \frac{\alpha \Delta T}{D} z$$

ϵ_{ij} longitudinal



$$du_x = \theta(z) \cdot dz \Rightarrow \int du_x = \int \theta(z) \cdot dz$$

$$u_x = \int \theta(z) \cdot dz + C(x_1)$$

$$\int_0^z du_x = \int_0^z \theta(z) \cdot dz = \int_0^z \frac{\alpha \Delta T}{D} \cdot z = \frac{\alpha \Delta T}{D} \cdot \frac{z^2}{2} \Big|_0^z = \frac{\alpha \Delta T}{D} \cdot \frac{z^2}{2}$$

$$u_x(z) - \underbrace{u_x(0)}_0 = \frac{\alpha \Delta T}{D} \cdot \frac{z^2}{2} \Rightarrow u_x(z) = \frac{\alpha \Delta T}{D} \cdot \frac{z^2}{2}$$

$$u_x(l) = 10^{-5} \frac{\text{cm}}{\text{mm}^\circ\text{C}} \cdot \frac{20^\circ\text{C}}{180\text{mm}} \cdot \frac{(4320\text{mm})^2}{2} = 10,368\text{mm}$$

